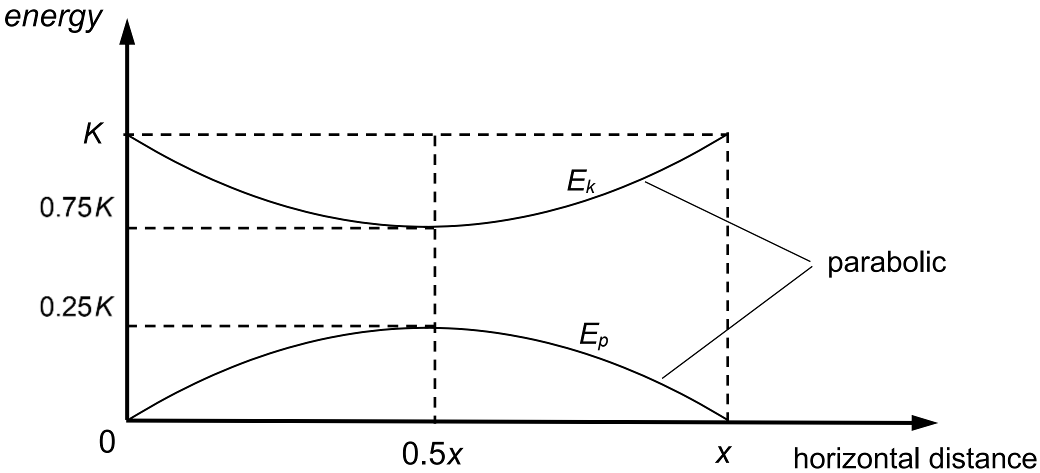
Q1			
(a)	(i)	Vertical component of the ball's initial velocity, $u_y = u \sin\theta$ $= 25 \sin 30°$ $= 12.5 \text{ m s}^{-1}$ $= 13 \text{ m s}^{-1}$	
	(ii)	Considering the vertical component motion, At maximum height, the vertical component of the ball's velocity will decrease to zero. Using $v_y^2 = u_y^2 + 2a_y s_y$ and taking upwards direction as positive, $0 = (12.5)^2 + 2(-9.81)s_y$ Vertical displacement (i.e. max height reached), $s_y = 7.96 = 8.0$ m	
	(iii)	Initial total energy = $K = \frac{1}{2}mu^2$ At maximum height of 8.0 m, ball's vertical component velocity, $v_y = 0$. Its velocity will be its horizontal component velocity, $v_x = u \cos 30° = \frac{\sqrt{3}}{2}u$ kinetic energy at 8.0 m = $\frac{1}{2}m\left(\frac{\sqrt{3}}{2}u\right)^2 = \frac{3}{4}\left(\frac{1}{2}mu^2\right) = 0.75K$ Gain in potential energy of system = Loss in kinetic energy of ball Potential energy at 8.0 m = $K - \frac{3}{4}K = 0.25K$ OR: Initial total energy at 8.0 m, $K = \frac{1}{2}mu^2 \Rightarrow m = \frac{2K}{u^2}$ $PE = mgh = \frac{2K}{u^2}gh = \frac{2(9.81)(8.0)}{(25)^2}K = 0.25K\,(2 \text{ s.f.})$ $KE = K - PE = K - 0.25K = 0.75K\,(2 \text{ d.p.})$	

(b)	**(i)**	 **Fig. 1.2** *Correct graph* E_p parabolic curve starts from zero, peaks at $K/4$ at the middle, ends at zero. E_k parabolic curve starts from K, decreases to $3K/4$ at the middle, ends at K. Correct labels, clear markings, symmetry	`